

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final Exam
 Duration: 1 Hour 40 Minutes

Semester: Spring 2023
 Full Marks: 40

CSE 221: Algorithms

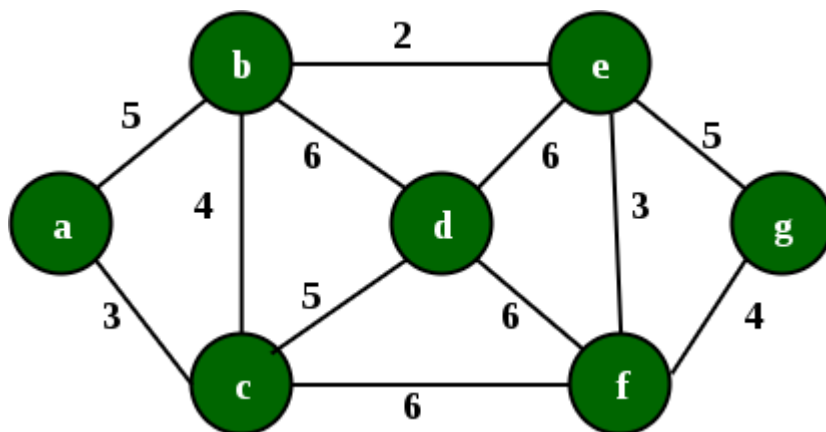
Answer the following questions.
 Figures in the right margin indicate marks.

Name:	ID:	Section:
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- 1 a. You work as an engineer for the Roads and Highways department in a district. The district has 7 thanas that are represented by the 7 vertices in the following graph. The edges between the vertices represent the roads that connect one thana with another and the weight of an edge represents the length of the road. 07
- CO2

A recent flood has totally damaged these roads and immediate repair work is needed. The cost to repair a road is proportionate to its length. However, your department does not have the budget to repair all the roads so you need to repair a subset of the roads. Which algorithm should you use here to find out the roads that need to be repaired to keep all the thanas connected to one another with the **minimum** possible cost?

Show the step by step simulation of the algorithm. You may pick any algorithm of your preference that you think will solve the problem.



- b. Problem Y is NP-complete. If A professor takes an instance of Y, converts it into an instance of the shortest path problem in 2^n time, and claims that Y is in P now, is the claim valid? 02
- CO3 **Explain** your answer.
- c. **Rank** the following three approximation algorithms from the worst to the best: 01
- CO3 2-approximation algorithm, 1.5-approximation algorithm, n-approximation algorithm

- 2 Two infamous thieves, Denver and Nairobi, planned to rob the famous Louvre Museum. Before the scene, they both agreed on the fact that none of them will break any item as all the items in the museum are too precious, and taking a fraction of any item won't sell on the black market. If it fits in the bag as a whole, they will take it, otherwise, leave it as it is.

Both of them arrived at the Royal Treasury with an empty knapsack weighing a total of **7 kg** each. Even though both thieves are experts in their fields, they take slightly different approaches. Denver believes he will use a Dynamic Programming Approach to rob the items in the most efficient manner possible. Nairobi, on the other hand, believes that if she chooses a Greedy Approach, she will make the most money.

The objects in the Royal Treasury Museum are listed below.

Objects	Diamond	Jewelry	Sculpture	Painting	Gold Crest
Profit (\$)	3	4	12	9	12
Weight (kg)	1	2	8	4	5

- CO2 a) **Simulate** your dynamic programming algorithm to find the maximum profit Denver can make. **05**
- CO2 b) From your memory/dp table, **find** out which items would he take to make this amount of profit? **02**
- CO3 c) Does Nairobi's belief remain valid after the robbery? **Prove** it. **03**

- 3 Following are the codes generated from a text for a Huffman tree construction.

<i>H</i> - 1000	<i>u</i> - 000	<i>e</i> - 011
<i>o</i> - 1001	<i>d</i> - 001	<i>l</i> - 110
<space> - 1010	<i>n</i> - 010	<i>t</i> - 111
<i>S</i> - 1011		

You are also given the following information:

- The frequency of each leaf node except *e*, *l*, and *t* is 1.
- The left and right child nodes of the root have frequencies 5 and 8 respectively.

Now answer the following questions.

- CO2 a) Suppose in a Huffman tree, the distances from the root to the pair of leaves denoting the letters *k* and *b* are 5 and 2 respectively, which letter between them is more frequent in the original text? Just **mention** the letter. **01**
- b) **Draw** the Huffman tree from the given coding table above. **03**
- c) Continuing on Q(b), what are the frequencies of *l*, and *t* in the original text? Just **mention** the frequencies. **02**

Your lazy friend, Tom, used a very simple encoding method for compressing a text file. The text file contains the string : *Sudden Hello*. He used a constant number of bits for encoding each of the *n* distinct characters in this text. The constant is $\text{ceil}(\log_2 n)$ in this case. On the other hand, you have compressed the text file using your own Huffman tree from Q(b).

- d) **Mention** how many bits each of the characters in Tom's encoding scheme contain. **01**
- e) **Compare** between the number of bits needed to decode the above-mentioned string using your friend's scheme and your Huffman tree from Q(b). **03**

Answer only one of 4, 4(or)

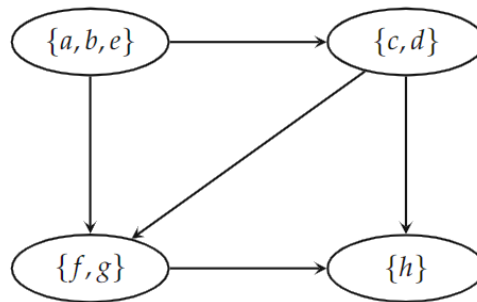
- 4 Your friend is trying to find the strongly connected components of a directed graph G . Recall that this involves first running depth-first search on G , and then running depth-first search on the reverse graph, G^R , using the decreasing order of finish time. Your friend has done this and found the following table. Here, $a, b, c, d, e, f, g, h, i$ are the vertices of G .

	a	b	c	d	e	f	g	h	i
$\text{finish}(G)$	6	4	8	18	5	14	17	16	15
$\text{finish}(G^R)$	18	17	12	10	16	6	8	9	4

Now answer the following questions.

CO2

- a) **Find** a vertex in G that is guaranteed to be contained in a source connected component. **01**
 Recall that a source connected component in an SCC DAG is a node that has indegree zero.
 For example, in the following SCC DAG, $\{a, b, e\}$ is a source connected component.

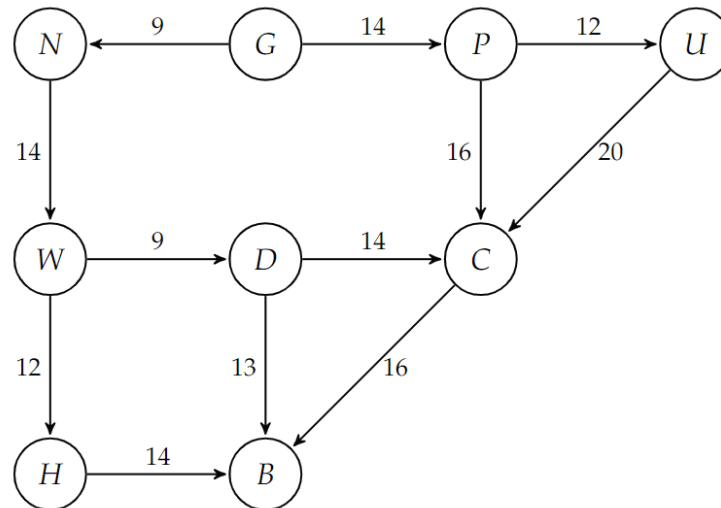


This image is just an example to show how SCC can convert a graph into DAG, and the source connected component

- b) Let x be your answer to (a). **Find** all the vertices in the strongly connected component that contains x . **05**
- c) If the vertices you found in (b) are deleted from G , which set of vertices is guaranteed to be a source connected component of the resulting graph? **Write** the list of those vertices. **01**
- d) How many strongly connected components does G have? Just **write** the answer. **01**
- e) **Draw** the DAG of the strongly connected components of G . There is more than one correct answer, anyone of those will be accepted. **02**

Question 4 (or) is on the next page.

4 or Consider the network of roads connecting a set of cities given by the graph below.



CO2

- Simulate** Dijkstra's algorithm to find the weight of the shortest path from G to B .
- There is a proposal to add one new road to this network, and there is a list of five pairs of cities between which the new road can be built. Each such potential road has an associated length. As a designer for the public works department, you are asked to determine the road whose addition to the existing network would result in the maximum decrease in the distance from G to B . One way to do this would be to add each of the five edges separately, run Dijkstra's algorithm five different times, and then compare the results. But can you do this by invoking Dijkstra's algorithm only two times (and possibly doing a linear amount of extra work)?

08

02

Present your idea with a pseudo-code/flowchart/step by step instruction/algorithm explanation.

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Mid Semester Exam
 Duration: 1 Hour 40 Minutes

Semester: Spring 2023
 Full Marks: 40

CSE 221: Algorithms

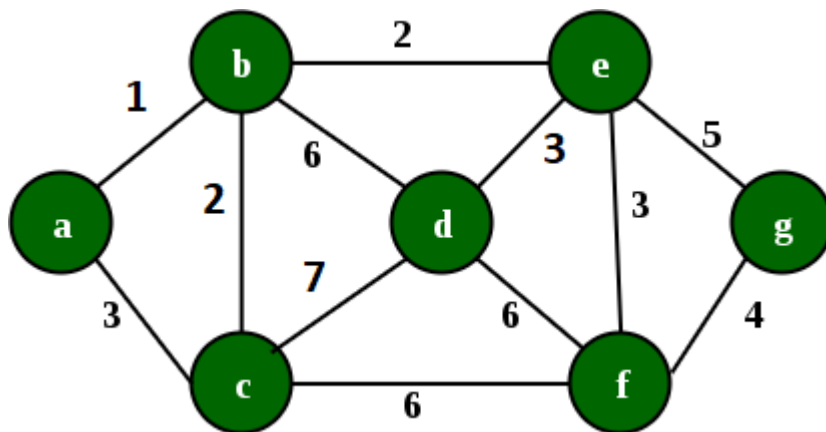
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- 1 a. You work as an engineer for the Roads and Highways department in a district. The district has 7 thanas that are represented by the 7 vertices in the following graph. The edges between the vertices represent the roads that connect one thana with another and the weight of an edge represents the length of the road. 07
- CO2

A recent flood has totally damaged these roads and immediate repair work is needed. The cost to repair a road is proportionate to its length. However, your department does not have the budget to repair all the roads so you need to repair a subset of the roads. Which algorithm should you use here to find out the roads that need to be repaired to keep all the thanas connected to one another with the **minimum** possible cost?

Show the step by step simulation of the algorithm. You may pick any algorithm of your preference that you think will solve the problem.



- b. Problem Y is NP-complete. If A professor takes an instance of Y, converts it into an instance of the shortest path problem in n^3 time, and claims that Y is in P now, is the claim valid? 02
- CO3
- Explain** your answer.
- c. **Rank** the following three approximation algorithms from the worst to the best: 01
- CO3 n-approximation algorithm, 2-approximation algorithm, 5-approximation algorithm

- 2 You are Monco, a video game character, you have just stepped into level 2. This game requires you to acquire necessity items to sustain through a level and this is done by trading reward points from the previous level. From level 1 you have gained 9,000 points. Below is a list of items you can get by trading your reward points. Each item has certain properties but be wise to pick the ones which will maximize your sustainability by maximizing mission impact.

	Gold Bar	Pain Killer	Waterproof Boots	Sun glasses	Energy Bar	Night Vision
Trade Price (points)	8000	2000	2000	2000	1000	3000
Mission Impact	1	5	3	2	2	4

- CO2 a) **Simulate** your chosen algorithm on the below items to find your picked items. 07
CO3 b) If you were to apply brute force, what would be the time complexity and why? **Explain.** 03
Assume that points you have gained are denoted as P, and the number of items is N.

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<i>J</i> - 1000	<i>n</i> - 000	<i>l</i> - 011
<i>o</i> - 1001	<i>h</i> - 001	<i>e</i> - 110
<space> - 1010	<i>m</i> - 010	<i>c</i> - 111
<i>W</i> - 1011		

You are also given the following information:

- The frequency of each leaf node except *e*, *l*, and *c* is 1.
- The left and right child nodes of the root have frequencies 5 and 8 respectively.

Now answer the following questions.

- CO2 a) Suppose in a Huffman tree, the distances from the root to the pair of leaves denoting the letters *m* and *e* are 3 and 5 respectively, which letter between them is more frequent in the original text? Just **mention** the letter. 01
b) **Draw** the Huffman tree from the given coding table above. 03
c) Continuing on Q(b), what are the frequencies of *e*, and *c* in the original text? Just **mention** the frequencies. 02

Your lazy friend, Tom, used a very simple encoding method for compressing a text file. The text file contains the string : *Welcome John*. He used a constant number of bits for encoding each of the *n* distinct characters in this text. The constant is $\lceil \log_2 n \rceil$ in this case. On the other hand, you have compressed the text file using your own huffman tree from Q(b).

- d) **Mention** how many bits each of the characters in Tom's encoding scheme contain. 01
e) **Compare** between the number of bits needed to decode the above-mentioned string using your friend's scheme and your Huffman tree from Q(b). 03

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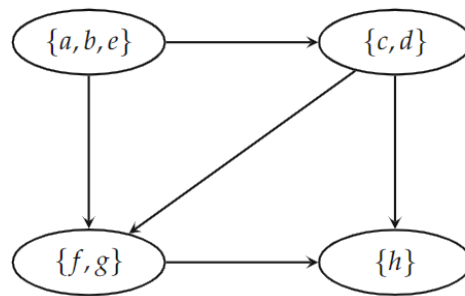
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	a	b	c	d	e	f	g	h	i
$\text{finish}(G)$	7	8	9	14	6	18	13	17	16
$\text{finish}(G^R)$	16	18	12	6	17	10	7	8	9

Now answer the following questions.

CO2

- a) **Find** a vertex in G that is guaranteed to be contained in a source connected component. Recall that a source connected component in an SCC DAG is a node that has indegree zero. For example, in the following SCC DAG, $\{a, b, e\}$ is a source connected component. 01

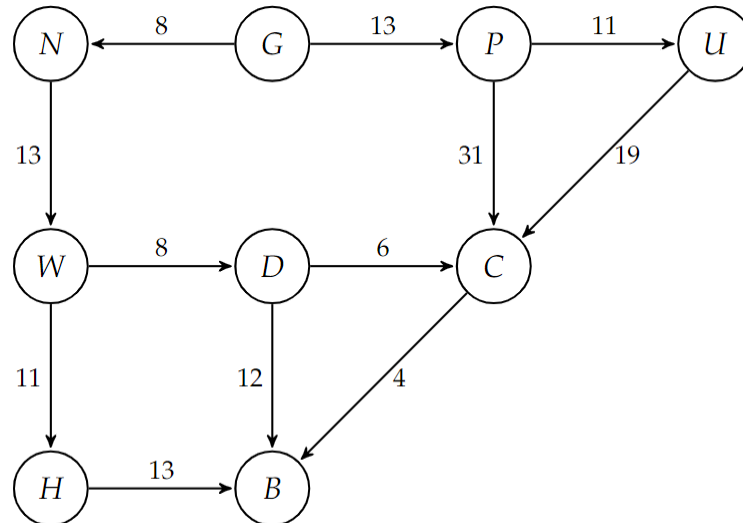


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- b) Let x be your answer to (a). **Find** all the vertices in the strongly connected component that contains x . 05
- c) If the vertices you found in (b) are deleted from G , which set of vertices is guaranteed to be a source connected component of the resulting graph? **Write** the list of those vertices. 01
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CO2

- Simulate** Dijkstra's algorithm to find the weight of the shortest path from G to B .
- There is a proposal to add one new road to this network, and there is a list of five pairs of cities between which the new road can be built. Each such potential road has an associated length. As a designer for the public works department, you are asked to determine the road whose addition to the existing network would result in the maximum decrease in the distance from G to B . One way to do this would be to add each of the five edges separately, run Dijkstra's algorithm five different times, and then compare the results. But can you do this by invoking Dijkstra's algorithm only two times (and possibly doing a linear amount of extra work)?

Present your idea with a pseudo-code/flowchart/step by step instruction/algorithm explanation.

08

02

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Final Exam
 Duration: 1 Hour 15 Minutes

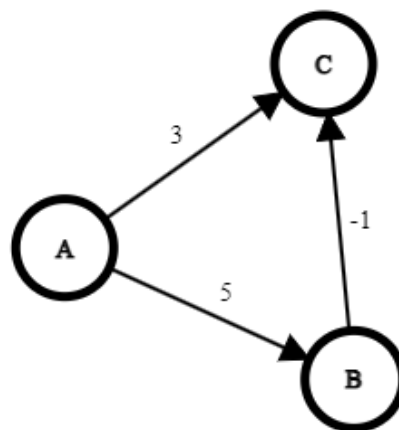
Semester: Spring 2024
 Full Marks: 30

CSE 221: Algorithms

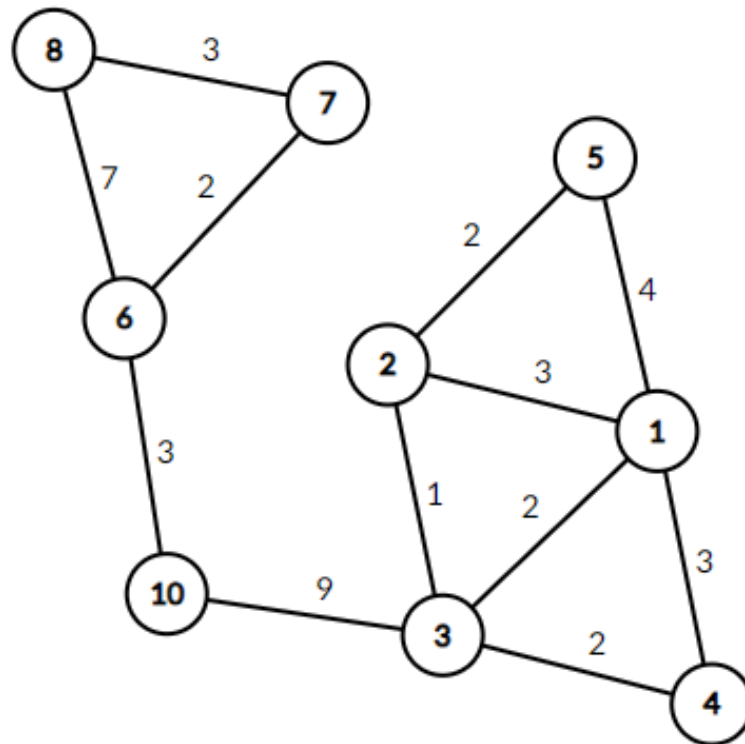
Answer the following questions.
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Name:	ID:	Section:
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- 1** Professor Bumstead needs to figure out a proper order of clothing when getting dressed. He notes down some of the dependencies which describe which garments he needs to put on before others. For example, Prof Bumstead needs to put on his shirt before he puts on his tie or his belt. He wrote down the following:
1. undershorts, watch and socks are independent
 2. pants need to be put on after undershorts
 3. shoes need to be put on after pants, undershorts and socks
 4. belt needs to be put on after pants and shirt
 5. tie needs to be put on after shirt
 6. jacket needs to be put on after tie and belt
- CO1** a. **Write** down the name of an appropriate algorithm to find the proper order of clothing items. **01**
- CO1** b. **Simulate** the algorithm by drawing a proper graph and determining the correct order of clothing items. If you can choose among multiple items, choose the alphabetically smallest one. **05**
 For example, between “shirt” and “shoes”, s comes before o, so choose “shirt”.
- CO1** c. Based on the clothing dependencies provided by Professor Bumstead, **identify** if there are any strongly connected components in the graph representation using an algorithm. **04**
 Demonstrate the steps of the algorithm you are applying or justify your answer in brief.
- 2 a.** You have this graph where there is a negative weight on one edge. **Explain** why Dijkstra’s shortest path algorithm will find correct answers on this graph given node A as the source. Then **propose** a change in the graph to make the result incorrect. **02**
- CO2**



- b. Someone mistakenly used Max-heap data structure instead of Min-heap to create a priority queue. So a pop/remove operation on the queue brings out the highest value element instead of the lowest one. Then this priority queue was used to implement Dijkstra's algorithm. Should it still work on any input graph? **Explain** your answer with proper reasoning and/or example. Assume that there will be no negative edges in the input. **03**
- c. Construct the minimum spanning tree from the graph below. Present your work with proper details/data structures. Finally, **draw** the MST and **write** the total cost of that tree. **05**



- 3 Wye is a CSE BracU Student. The pre-advising week is approaching, and he is trying to find the most suitable course combination for next semester. As per his previous academic results, his credit limit **has been set to 9**. Different courses have different credits. He came up with values for these courses by consulting with students who have already taken them. He wants to select the most valued courses. Courses, credits and values are provided below:

Course	PHY112	CSE330	CSE331	CSE370	CSE499
Credit	1	2	3	4	5
Value	9	7	15	12	16

- CO1 a. Wye wants to try all possible combinations to find the best one. Help him by simulating a suitable dynamic programming approach. **Show** your work in detail, then **write** down the selected courses. **04**
- CO1 b. Zed, a friend of Wye, adopted a slightly different approach and considered the courses based on their per-credit value. **Determine** whether Zed could select more valued courses than Wye. **02**
- CO1 Note that, selecting a course still means taking the whole, with full credits.
- c. If Zed is successful, he will send a message **Successful**; otherwise, **Unsuccessful**. Instead of fixed-length coding, he is willing to send the message using Huffman Coding. **Show** simulation of Huffman coding to encode his message. **04**

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CSE 221: Algorithms

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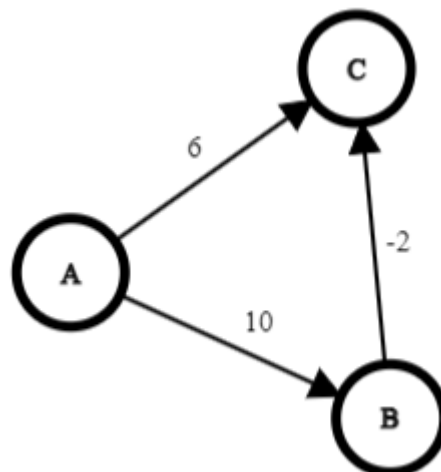
Name:	ID:	Section:
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- 1 Mr. Smith, a carpenter, needs to figure out a proper order of assembling items when making a bookshelf. He notes down some of the dependencies which describe which task he needs to complete before others. For example, he needs to create the frame before attaching the shelves. He wrote down the following:

1. Frame, Paint and Shelf-stand are independent
2. Back-panel needs to be installed after Frame
3. Drawer needs to be installed after Paint, Back-Panel and Shelves
4. Shelves need to be installed after Back-panel
5. Screws need to be installed after Shelves and Back-panel
6. Books need to be installed after Drawers
7. Knobs need to be installed after Drawers

- CO1 a. **Write** down the name of an appropriate algorithm to find the proper order of tasks. **01**
- CO1 b. **Simulate** the algorithm by drawing a proper graph and determining the correct order of tasks. If you can choose among multiple items, choose the alphabetically smallest one. For example, between “screws” and “shelves”, c comes before h, so choose “screws”. **05**
- CO1 c. Based on the dependencies provided by Mr. Smith, **identify** if there are any strongly connected components in the graph representation using an algorithm. **04**
 Demonstrate the steps of the algorithm you are applying or justify your answer in brief.

- 2 a. You have this graph where there is a negative weight on one edge. **Explain** why Dijkstra’s shortest path algorithm will find correct answers on this graph given node A as the source. Then **propose** a change in the graph to make the result incorrect. **02**

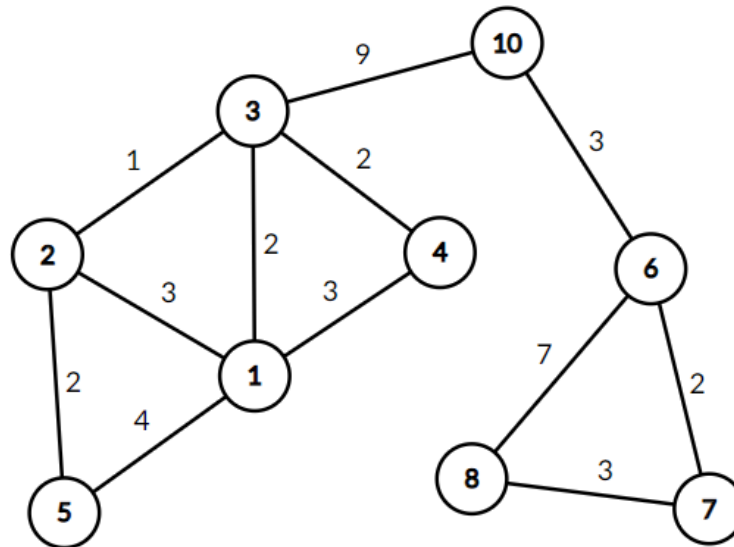


- b.** Dijkstra's algorithm inserts all vertices in a priority queue/Min-heap at the beginning and then repeatedly extracts the vertex with minimum distance. However, someone proposed the following changes: **03**
- CO2**

1. At the beginning, it inserts only the source vertex in the priority queue.
2. Whenever the distance of a vertex is updated, it then inserts that vertex into the queue.

Should the time complexity of the algorithm be same as before? **Explain** your answer with proper reasoning and/or example. Assume that there will be no negative edges in the input.

- c.** Construct the minimum spanning tree from the graph below. Present your work with proper details/data structures. Finally, **draw** the MST and **write** the total cost of that tree. **05**
- CO1**



- 3** Wye is a CSE BracU Student. The pre-advising week is approaching, and he is trying to find the most suitable course combination for next semester. As per his previous academic results, his credit limit **has been set to 9**. Different courses have different credits. He came up with values for these courses by consulting with students who have already taken them. He wants to select the most valued courses.

Courses, credits and values are provided below:

Course	PHY112	CSE330	CSE331	CSE370	CSE499
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- CO1** a. Wye wants to try all possible combinations to find the best one. Help him by simulating a suitable dynamic programming approach. **Show** your work in detail, then **write** down the selected courses. **04**
- CO1** b. Zed, a friend of Wye, adopted a slightly different approach and considered the courses based on their per-credit value. **Determine** whether Zed could select more valued courses than Wye. **02**
Note that, selecting a course still means taking the whole, with full credits.
- CO1** c. If Zed is successful, he will send a message **Successful**; otherwise, **Unsuccessful**. Instead of fixed-length coding, he is willing to send the message using Huffman Coding. **Show** simulation of Huffman coding to encode his message. **04**

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Semester: Spring 2025
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CSE 221: Algorithms

Answer the following questions.
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Name:	ID:	Section:
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- 1 a. For a directed graph, does the number of back edges equal the number of cycles present? **01**
CO1 **Explain** briefly.
- b. Given a directed graph $G = (V, E)$ represented with an adjacency list, **explain** how you can **01**
CO1 find whether the graph is a directed acyclic graph (DAG) or not. Your method must be as asymptotically efficient as possible.

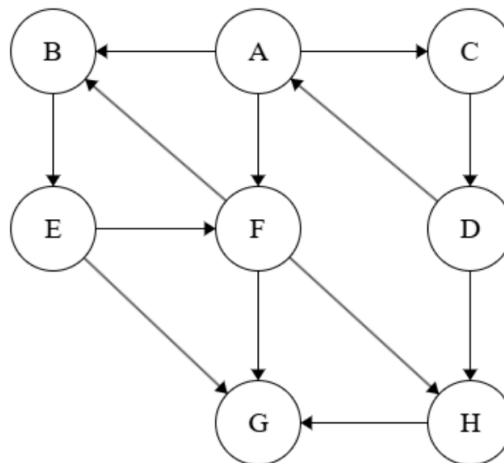


Figure 1: Q1(b)

- c. **Simulate** Kosaraju's algorithm on the given graph Q1(b) to find the strongly connected **04**
CO1 components (SCCs). Label each SCC with the set of nodes it contains. While performing graph traversal, use node A as the starting node and break ties in alphabetical order.
- d. Now, treat each SCC identified in Q1(c) as a single "super node." In this new graph, every **01**
CO1 edge from the original graph either becomes internal (and thus disappears, since it stays within an SCC) or becomes an edge connecting two different SCCs. This transformation results in a Directed Acyclic Graph (DAG). **Draw** this DAG.
- e. Given a set of SCCs, what is the maximum number of new SCCs that can appear if we **01**
CO1 delete one edge from the DAG of these SCCs?

Consider the following algorithm:

- Step 1: Perform DFS on the given graph. For each node, keep track of the finishing time (the time when the algorithm finishes exploring all descendants of the node).
- Step 2: Sort the nodes in ascending order of the finish time.
- Step 3: Perform DFS in the sorted order of the nodes. For each DFS traversal, include each reachable node as one SCC.

f. **CO2** **Prove** that the given algorithm is incorrect by providing a graph as a counterexample, on which the algorithm will fail to find the correct set of SCCs. **02**

2 The country of Technovia is setting up an emergency grid that connects 8 major cities: A (Capital), B, C, D, E, F, G, H. These cities are connected by one-way roads. Each road has a traffic weight which is used to estimate the delivery delay due to congestion. Medical aid and personnel must be delivered from the capital (A) to all other cities. The map of the country is given below:

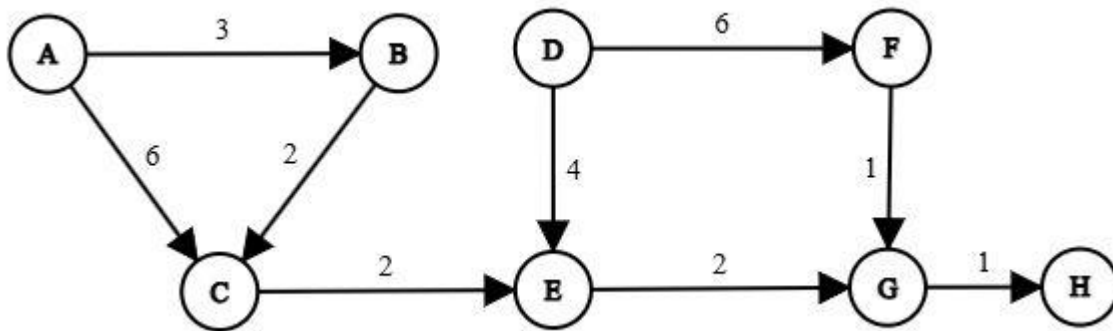


Figure 2: Q2.

a. **CO1** Suppose you, an algorithm enthusiast, have been assigned to determine the minimum delivery time from capital A to all other cities. **04**

1. Which SSSP (single source shortest path) algorithm would you prefer to solve the problem?
2. **Simulate** your mentioned algorithm with the necessary data structures to find the minimum delivery times from the capital to all other cities. Travel times are equal to the traffic weights of the roads.

b. **CO3** Continuing from the given scenario, the Ministry of Transport enforces that while delivering medical resources, after using any high-traffic road (traffic weight > 4), the next road used in the delivery path must have a traffic weight ≤ 4 . This is to prevent back-to-back gridlock. **02**

Propose a modification to your algorithm of Q2(a) to determine the minimum delivery time from capital A to all other cities while strictly following the additional constraint of traffic weights. **Present** your idea using programmable code/pseudocode/step-by-step instructions.

c. Now the Ministry wants to design a minimum-traffic infrastructure plan that connects all the cities (an MST of the graph). Will Kruskal's or Prim's MST finding algorithm work properly for the given one-way roads? If yes, **justify** your answer. Otherwise, **propose** necessary modifications to the graph so that the algorithm works. **02**

d. While finding the MST of the graph, we want to enforce a rule that '*No city can have more than two (02) roads*'. **02**

Propose a modification to the MST algorithm (either Kruskal's or Prim's) to accommodate this requirement. **Present** your idea using programmable code/pseudocode/step-by-step instructions.

3 The final exam has just ended, and you are on a well-deserved trip to a remote area. During your journey, you have collected different types of digital content that you want to share with your friends and family. Each *Content Type* has a *Total Duration* and a *Value*. Total Duration measures the content's pre-compression size, while Value reflects its importance or meaning to the audience.

Due to bandwidth constraints, one of your friends suggested using Huffman encoding, a lossless compression technique that assigns shorter binary codes to more voluminous (i.e., longer duration) items.

Content Type	Total Duration (Seconds)	Value
A	13	20
B	11	14
C	9	6
D	8	17
E	6	11
F	5	7
G	4	8

a. Construct a Huffman Tree based on the given **Total Duration** to assign optimal binary codes to each content type. **04**

b. Calculate the Total Number of Bits (*duration * code length*) required to transmit the full content of each type. **03**

c. Again, due to bandwidth constraints, you are now trying to select which content to send and how much of each. You can shorten any content, but its value drops proportionally if you do so. Your goal is to choose the types and lengths of the contents that give you the highest total value without going over *30 seconds*. **Determine** which content to send and how long each part should be. **03**

Please Turn Over

- 4 a. String1: <Your ID>
CO1 String2: _____
- i. Create String2 in a way that the following constraints match. **Write** the string. **03**
- Length of String2 = 8 (eight)
 - Length of LCS(String1,String2) = 5 (five)
 - There are exactly three distinct LCS (of length = 5)
- ii. **Write** all three LCS strings (just write, no need to show any work). **01**
- b. String1: <Your ID>
CO1 String2: "20251905"
- iii. **Find** the length of the LCS of these two strings. You can use a memory table or a recursion tree. Show your work. **04**
- iv. Using your work from the previous question, **find** the LCS string. **02**

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 Full Marks: 40

CSE 221: Algorithms

Answer the following questions.
 Figures in the right margin indicate marks.

Name:	ID:	Section:
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- 1 a. For a directed graph, does the number of back edges equal the number of cycles present? **01**
CO1 **Explain** briefly.
- b. Given a directed graph $G = (V, E)$ represented with an adjacency list, **explain** how you can **01**
CO1 find whether the graph is a directed acyclic graph (DAG) or not. Your method must be as asymptotically efficient as possible.

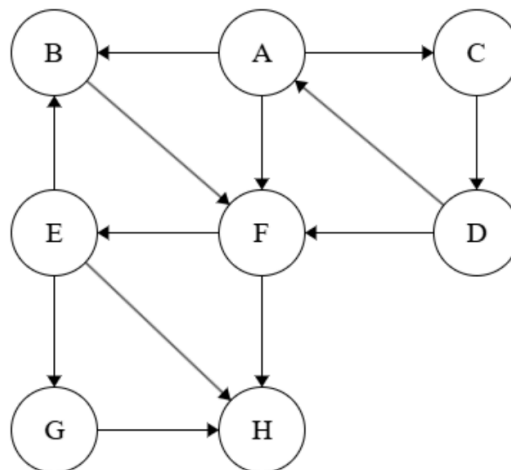


Figure 1: Q1(b)

- c. **Simulate** Kosaraju's algorithm on the given graph Q1(b) to find the strongly connected **04**
CO1 components (SCCs). Label each SCC with the set of nodes it contains. While performing graph traversal, use node A as the starting node and break ties in alphabetical order.
- d. Now, treat each SCC identified in Q1(c) as a single "super node." In this new graph, every **01**
CO1 edge from the original graph either becomes internal (and thus disappears, since it stays within an SCC) or becomes an edge connecting two different SCCs. This transformation results in a Directed Acyclic Graph (DAG). **Draw** this DAG.
- e. Given a set of SCCs, what is the maximum number of new SCCs that can appear if we **01**
CO1 delete one edge from the DAG of these SCCs?

Consider the following algorithm:

- Step 1: Perform DFS on the given graph. For each node, keep track of the finishing time (the time when the algorithm finishes exploring all descendants of the node).
- Step 2: Arrange the nodes in ascending order based on the finish time.
- Step 3: Perform DFS in the sorted order of the nodes. For each DFS traversal, include each reachable node as one SCC.

f. **CO2** **Prove** that the given algorithm is incorrect by providing a graph as a counterexample, on which the algorithm will fail to find the correct set of SCCs. **02**

- 2 The country of Technovia is setting up an emergency grid that connects 8 major cities: A (Capital), B, C, D, E, F, G, H. These cities are connected by one-way roads. Each road has a traffic weight which is used to estimate the delivery delay due to congestion. Medical aid and personnel must be delivered from the capital (A) to all other cities. The map of the country is given below:

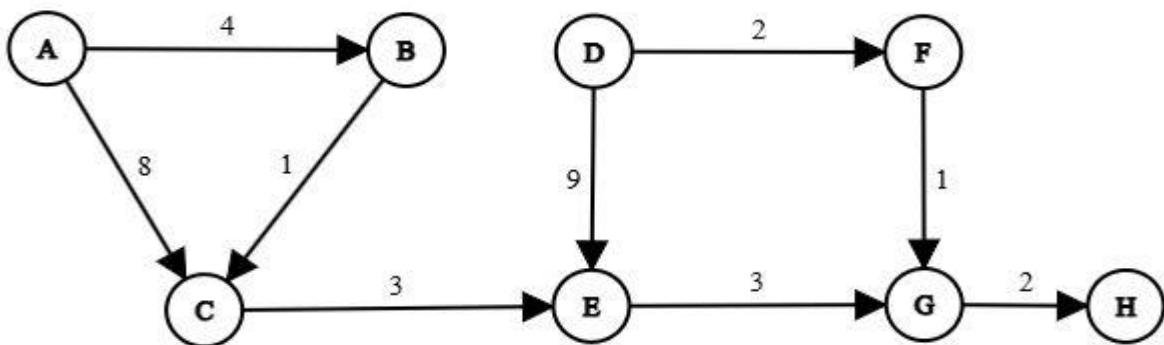


Figure 2: Q2

a. **CO1** Suppose you, an algorithm enthusiast, have been assigned to determine the minimum delivery time from capital A to all other cities. **04**

1. Which SSSP (single source shortest path) algorithm would you prefer to solve the problem?
2. **Simulate** your mentioned algorithm with the necessary data structures to find the minimum delivery times from the capital to all other cities. Travel times are equal to the traffic weights of the roads.

b. **CO3** Continuing from the given scenario, the Ministry of Transport enforces that while delivering medical resources, after using any high-traffic road (traffic weight > 5), the next road used in the delivery path must have a traffic weight ≤ 5 . This is to prevent back-to-back gridlock. **02**

Propose a modification to your algorithm of Q2(a) to determine the minimum delivery time from capital A to all other cities while strictly following the additional constraint of traffic weights. **Present** your idea using programmable code/pseudocode/step-by-step instructions.

c. Now the Ministry wants to design a minimum-traffic infrastructure plan that connects all the cities (an MST of the graph). Will Kruskal's or Prim's MST finding algorithm work properly for the given one-way roads? If yes, **justify** your answer. Otherwise, **propose** necessary modifications to the graph so that the algorithm works. **02**

d. While finding the MST of the graph, we want to enforce a rule that '*No city can have more than two (02) roads*'. **02**

CO3

Propose a modification to the MST algorithm (either Kruskal's or Prim's) to accommodate this requirement. **Present** your idea using programmable code/pseudocode/step-by-step instructions.

3 The final exam has just ended, and you are on a well-deserved trip to a remote area. During your journey, you have collected different types of digital content that you want to share with your friends and family. Each *Content Type* has a *Total Duration* and a *Value*. Total Duration measures the content's pre-compression size, while Value reflects its importance or meaning to the audience.

Due to bandwidth constraints, one of your friends suggested using Huffman encoding, a lossless compression technique that assigns shorter binary codes to more voluminous (i.e., longer duration) items.

Content Type	Total Duration (Seconds)	Value
A	14	21
B	12	15
C	10	7
D	9	18
E	7	12
F	6	8
G	5	7

a. Construct a Huffman Tree based on the given **Total Duration** to assign optimal binary codes to each content type. **04**

CO1

b. Calculate the Total Number of Bits (*duration * code length*) required to transmit the full content of each type. **03**

CO1

c. Again, due to bandwidth constraints, you are now trying to select which content to send and how much of each. You can shorten any content, but its value drops proportionally if you do so. Your goal is to choose the types and lengths of the contents that give you the highest total value without going over *35 seconds*. **Determine** which content to send and how long each part should be. **03**

CO1

Please Turn Over

4 a. String1: <Your ID>
CO1 String2: _____

- i. Create String2 in a way that the following constraints match. **Write** the string. **03**
 - o Length of String2 = 8 (eight)
 - o Length of LCS(String1,String2) = 5 (five)
 - o There are exactly three distinct LCS (of length = 5)
- ii. **Write** all three LCS strings (just write, no need to show any work). **01**

b. String1: <Your ID>
CO1 String2: "20251905"

- iii. **Find** the length of the LCS of these two strings. You can use a memory table or a recursion tree. Show your work. **04**
- iv. Using your work from the previous question, **find** the LCS string. **02**

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final Exam
 Duration: 1 Hour 45 Minutes

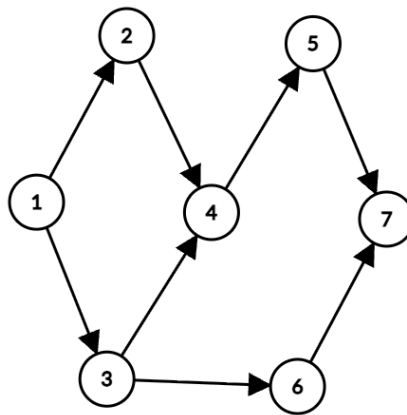
Semester: Spring 2026
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CSE 221: Algorithms

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- 1 CO1** A software build system manages modules $\{1, 2, 3, 4, 5, 6, 7\}$ where a directed edge $u \rightarrow v$ means module u must be compiled before module v . The dependency graph G has been provided below:



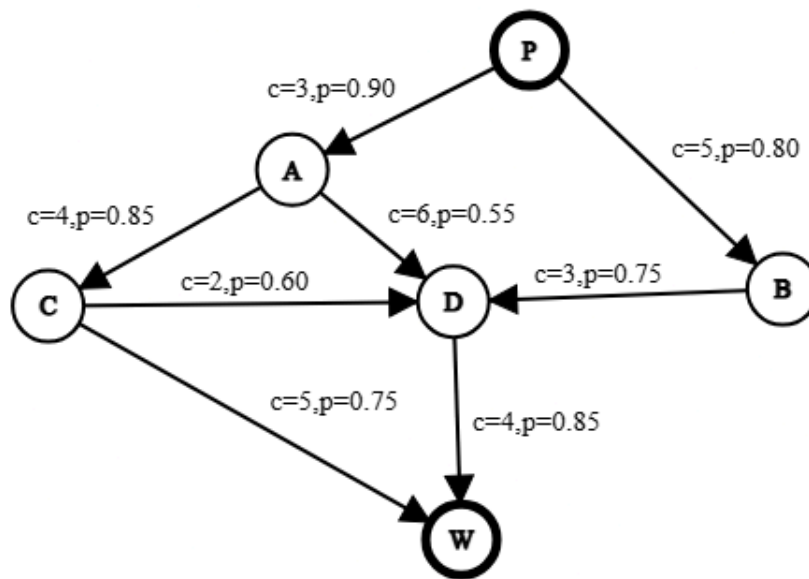
- a. **Simulate** DFS on G starting from module 1, visiting neighbors in **numerical order** (whenever there is a tie). **Record** discovery time $d[v]$ and finish time $f[v]$ for each module. **3**
- b. A valid compilation order requires every module to appear after all its dependencies. Using the finish times from Q(a), **determine** one such order. **3**
- c. A developer adds a new dependency $7 \rightarrow 3$, forming graph G' . **Identify** all cycles from G' . Just write your answer, no need to show any simulation. **2**
- d. Since G' contains cycles, a **topological sort no longer exists**. To fix the build order, merge all mutually dependent modules that form cycles into a single node. Modules not in any cycle remain as individual nodes. Using the cycles from Q(c), **list** the members of each merged group and **draw** the resulting dependency graph. **2**

2

A hospital's internal logistics system uses a courier robot to deliver critical medicine from the **Pharmacy (P)** to the **Intensive Care Ward (W)**. The hospital has 6 key locations (nodes) connected by corridors (edges). Each corridor has two attributes:

- $c(u, v)$ = traversal cost of moving the robot through the corridor
- $p(u, v)$ = probability the corridor is traversable at the time of delivery (some corridors periodically close for emergency procedures)

The hospital's safety protocol requires the robot to prefer routes with **lower traversal cost and higher traversability**, so that medicine can be delivered both efficiently and reliably.



- Which one is the optimal route to go from P to W? **Explain** why. 2
- CO1** Assume that,
 Total traversal cost = summation of the traversal costs (c) of all corridors in the route
 Total traversability = product of the probabilities (p) of all corridors in the route
- We want to convert the two variables into a single one called **effective cost, $w(u, v)$** so that a shortest path algorithm can be applied. **Define** a formula for $w(u, v)$ using both $c(u, v)$ and $p(u, v)$; such that w becomes smaller when c is low and p is high. 2
- CO3**
- Continuing from above, **simulate** a shortest path algorithm to find the route from **Pharmacy (P)** to **Intensive Care Ward (W)** with the minimum total effective cost. Show your steps clearly and **report** the final path with its total effective cost. 4
- CO1**
- "Minimizing the effective cost $w(u, v)$ does not always guarantee a path with both low total cost and high overall traversability."
Verify the statement above by providing either an example graph or a logical explanation. 2
- CO2**

3

Consider that, you are a scientist from a renowned technology company and you are working with data compression algorithms. Now you want to use Huffman algorithm for the data given below:

Character	Frequency
A	2
B	3
C	4
D	6
E	7
F	8
G	10

a. **CO1** Create the Huffman Tree to encode the characters and write the codeword for the characters. **5**

b. **CO2** Another friend tried to implement the same algorithm but made an error. The codewords s/he found for characters are given below: **3**

Character	Codeword
A	000000
B	000001
C	00001
D	0001
E	001
F	01
G	1

Explain where s/he made the mistake and what can be the consequences of this mistake.

c. **CO3** Suppose you have a storage limit of exactly 11 bits. You want to store as many unique characters as possible to maximize the variety of your data, but you must prioritize characters that appear most frequently in your dataset to ensure the most common data is preserved. Using a Greedy approach, **identify** which characters will be stored within the 11-bit limit. **2**

- 4 CO1** We are solving the 0/1 Knapsack problem using dynamic programming. We have a knapsack of capacity $W = 5$ and $n = 4$ items with the following weights and values:

Item	Weight	Value
1	x	3
2	3	y
3	1	2
4	4	5

We fill a 2D table K where $K[i][w]$ represents the maximum value achievable using the first i items with knapsack capacity w . However, some entries are masked using symbols as shown below:

	w=0	w=1	w=2	w=3	w=4	w=5
i=0	0	0	0	0	0	0
i=1	0	0	3	3	3	3
i=2	0	0	3	4	4	7
i=3	0	2	3	♥	7	♣
i=4	0	★	3	♠	◆	☆

- Using the DP table K , **determine** the weight of **Item#1 (x)** and value of **Item#2 (y)**. 2
- $K[2][4] = 4$ but $K[2][5] = 7$. How is it possible to achieve value 7 at capacity 5 using only items 1 and 2? **Explain**. 2
- Fill** up the dp table by determining the missing values for ♥, ♣, ★, ♠, ◆, and ☆. 3
- After completing the table, **how** would you trace back through K to determine which items are included in the optimal solution for $W = 5$? **Simulate** the traceback procedure in your script using the table from Q(c) and apply it to find the actual items chosen. 3

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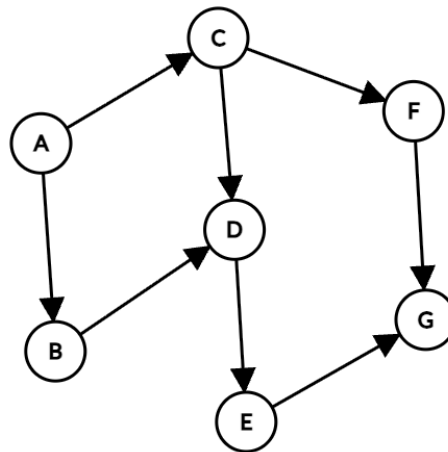
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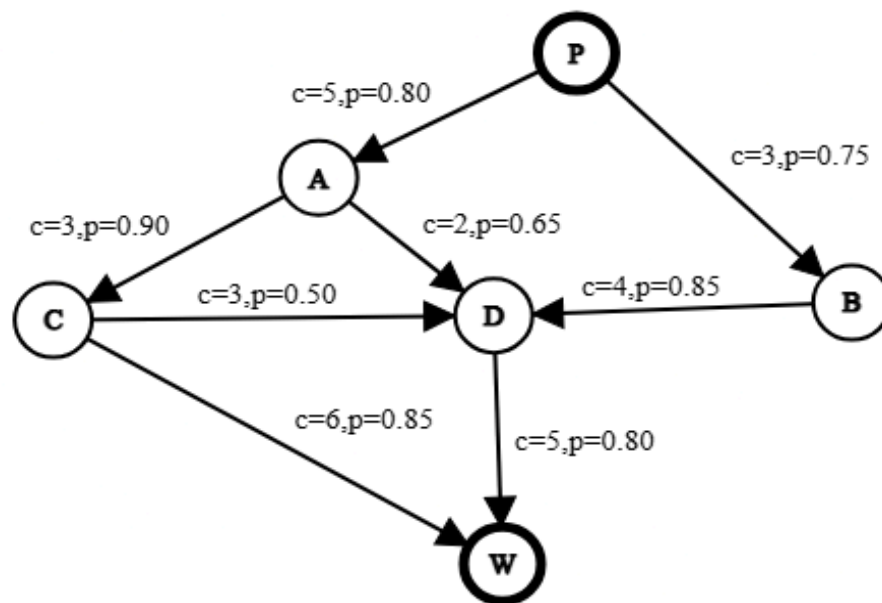
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